Vision Transformers Excel in Chest X-Ray Analysis

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***Abstract*—In this study, we evaluate multiple benchmark model architectures, including Vision Transformers (ViT), for multi-label classification of chest X-ray (CXR) images using the NIH CXR 14 dataset—a challenging benchmark for medical image analysis with annotated radiographs covering a wide range of diseases. Our results show that ViT outperforms DenseNet in terms of accuracy and Area Under the Curve (AUC) for most pathologies. For example, the AUC for Cardiomegaly is 0.9150 with ViT compared to 0.9126 with DenseNet, and for Emphysema, the AUC is 0.9380 with ViT versus 0.9360 with DenseNet. This work highlights the trade-offs between the localized feature extraction of Convolutional Neural Networks (CNNs) and the global contextual modeling of transformers, demonstrating the potential of both architectures in medical imaging tasks.**

***Index Terms*—Vision Transformer, CNN, Chest X-ray**

I. Introduction

Deep Neural Networks (DNNs) have transformed all domains, and medical image analysis is no exception [1], which includes diagnosis of pulmonary conditions like Lung Cancer, Pneumonia, and Tuberculosis. To be precise, Convolutional Neural Networks (CNNs), a key subset, excel in feature extraction, driving state-of-the-art classification performance [2]–[4]. Vision Transformers (ViTs) have emerged as a strong alternative to CNNs, capturing global context through self-attention rather than convolution. By segmenting images into patches, ViTs excel in tasks requiring long-range dependencies for accurate classification.

In this paper, we focus on the Chest X-Ray (CXR) 14 dataset [5], a comprehensive collection of over 100,000 labeled CXRs spanning 14 disease categories. We conduct a comparative analysis between DenseNet, a state-of-the-art CNN-based architecture, and ViT for disease classification. DenseNet leverages dense connections between layers to improve feature propagation and model efficiency, while ViT bring a novel approach to image processing by explicitly capturing relationships across the entire image. Both DenseNet and ViT models are trained on the CXR 14 dataset, and their performances are evaluated using various classification metrics, including accuracy, precision, recall, and F1-score. Our findings demonstrate that while DNNs, particularly DenseNet, remain robust for medical image classification, ViT outperforms DenseNet in capturing global dependencies. This leads to superior performance in automatic disease classification from CXRs.

Code Repository: https://github.com/2ai-lab/VIT-CXR

The availability of large datasets like CheXpert [6], MIMIC-CXR [7], and PadChest [8] has significantly advanced deep learning for CXR disease classification. CheXNet [9] first demonstrated CNNs' potential in pneumonia detection using DenseNet, a widely adopted architecture known for efficient feature reuse and gradient flow [10].

ViT, which treats images as patch sequences and leverage self-attention, offers superior global context modeling, proving effective in medical imaging tasks like classification and segmentation [11], [12]. Hybrid CNN-ViT models combine CNNs' localized feature extraction with transformers' global reasoning, as seen in the works of Zhou et al. [13] and Xie and Zhang [14], achieving state-of-the-art performance. Due to ViT's high computational cost, efficient variants like LightViT [15] and MobileViT [16] have been developed for resource-limited environments. Transformers also enhance multi-task learning frameworks, as shown by Park et al. [17], integrating classification and localization for improved clinical efficiency.

This study extends prior research by evaluating ViT and DenseNet on the CXR 14 dataset, providing insights into their suitability for automated disease diagnosis in clinical settings.

II. Revisiting ViT

This section outlines the fundamentals of ViT and its comparative evaluation against benchmark architectures. We then analyze and justify their selection for our study.

A. Context

ViT has gained prominence for its ability to model long-range dependencies in images by leveraging self-attention mechanisms. CNNs process information hierarchically through localized receptive fields. ViT divides an image into patches and processes them sequentially through a transformer encoder. This allows ViT to capture global contextual relationships more effectively than traditional CNNs.

2025 IEEE Conference on Artificial Intelligence (CAI) | 979-8-3315-2400-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/CAI64502.2025.00090

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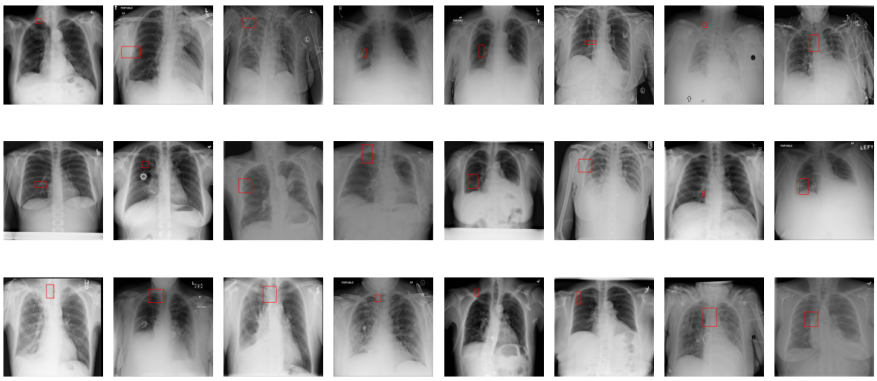


Fig. 1: Overview of the CXR 14 dataset, illustrating sample images across various disease categories.

Among CNN-based models, DenseNet has been widely used for medical image analysis due to its efficient feature reuse and gradient flow. It employs dense connectivity patterns, ensuring that each layer receives direct gradients from preceding layers, leading to enhanced feature propagation. However, its reliance on convolutional operations limits its ability to capture global dependencies.

B. Why ViT?

While models such as DenseNet, ResNet, and EfficientNet have demonstrated strong performance in medical imaging, ViT excels in cases requiring broader contextual understanding. The key reasons for selecting ViT in our study include:

- *Long-range dependencies:* ViT leverages self-attention mechanisms to model complex interactions across the entire image, which is crucial for detecting diffuse or large-scale abnormalities such as cardiomegaly and emphysema.
- *Adaptability to different pathologies:* ViT outperforms CNNs in diseases where subtle, distributed patterns exist across the image rather than localized features.
- *Competitive performance:* As shown in Table **??**, ViT consistently achieved higher AUC values across multiple pathologies while maintaining robust generalization.
- *Scalability:* Unlike CNNs, which require extensive hyperparameter tuning for different image resolutions, ViT exhibits strong scalability without major architectural modifications.

Our benchmark comparison (*ref.* Table IV) further highlights that ViT delivers superior classification accuracy in pathology detection, particularly in cases where understanding the broader image context is critical.

C. Benchmark evaluation summary

Table I provides a detailed comparison of ViT and CNN-based architectures, including DenseNet121, ResNet50, EfficientNetB7, and Xception. The comparison highlights fundamental differences in their structural design, computational efficiency, and ability to capture spatial features in medical imaging tasks.

a) *Input processing and feature representation:* All models utilize 224×224 pixel input images with standard normalization in the [0,1] range. While CNN-based architectures process images through hierarchical convolutional layers, ViT decomposes images into patch embeddings and processes them sequentially via self-attention mechanisms. This fundamental difference enables ViT to capture global dependencies across the entire image, whereas CNNs rely on local receptive fields to extract spatial features.

b) *Computational complexity:* ViT demands significantly higher computational resources compared to CNNs due to its multi-head self-attention mechanism, which scales quadratically with the number of patches. In contrast, models like ResNet50 and DenseNet121 use weight-sharing convolutional filters, leading to lower memory and compute requirements. EfficientNetB7, while achieving improved performance through compound scaling, is also more resource-intensive than standard CNNs but remains computationally more efficient than ViT.

c) *Strengths and use cases:* CNN architectures such as DenseNet121 and ResNet50 excel at detecting localized features (e.g., nodules, lesions) due to their ability to preserve spatial hierarchies. Xception, with its depthwise separable convolutions, enhances feature extraction while reducing computational overhead. EfficientNetB7, leveraging compound scaling, balances depth, width, and resolution to achieve high accuracy while maintaining efficiency. On the other hand, ViT outperforms CNNs in capturing long-range dependencies, making it particularly useful for identifying diffuse or complex patterns (e.g., cardiomegaly, emphysema).

d) *Suitability for medical image analysis:* While CNNs remain the preferred choice for computational efficiency

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TABLE I: Detailed comparison of ViT and other CNN architectures

Parameter	ViT	DenseNet	ResNet50	EfficientNetB7	Xception
Input Size	224×224	224×224	224×224	224×224	224×224
Norm Range	[0,1]	[0,1]	[-1,1]	[0,1]	[-1,1]
Augmentation	Random flips, zooms	Random flips, zooms	Crops, jitter	AutoAugment, erasing	Shifts, distortions
Loss	BCE	BCE	BCE	BCE	BCE
Optimizer	Adam (1e-4)	Adam (1e-4)	SGD (lr - 0.01)	RMSProp (0.001)	Adam (1e-4)
LR Decay	0.1 (5 epochs)	0.1 (5 epochs)	0.1 (30 epochs)	Scheduled	Adaptive
Batch Size	32	32	32	32	32
Training	50 epochs	50 epochs	100 epochs	300 epochs	50 epochs
Efficiency	High demand	Low cost	Moderate	Optimized	High computation
Strengths	Captures global info	Excels in local features	Strong generalization	High accuracy	Efficient convolutions

and localized feature extraction, ViT offers superior performance in cases requiring global contextual understanding. EfficientNetB7, with its optimized scaling strategy, provides a strong alternative by achieving state-of-the-art accuracy with lower computational cost compared to deeper CNNs. However, deploying ViT in real-world settings necessitates efficient memory management and potentially hybrid architectures, combining CNN feature extraction with transformer-based attention mechanisms.

In summary, the comparison in Table I underscores the strengths and trade-offs of each architecture. The choice between ViT and CNN-based models depends on the specific imaging task, computational resources, and the scale of available training data.

III. Implementation and results

The implementation of ViT and DenseNet was carried out using the PyTorch deep learning framework. Both models were fine-tuned on the NIH CXR 14 dataset. Below, we outline the implementation details and configurations used for each model:

A. Training and optimization

To ensure a fair comparison between the ViT and DenseNet models, both were trained under identical conditions. The key configurations for the training pipeline are summarized in Table I.

The dataset preprocessing steps included resizing the input images to 224×224 pixels to maintain consistency across both the ViT and DenseNet models. Pixel values were normalized to the range [0, 1], which stabilized the gradient flow during training. To improve model generalization and reduce overfitting, data augmentation techniques such as random flips, rotations, and zooms were employed.

Both models utilized the Adam optimizer with an initial learning rate of 1×10^{-4}. The learning rate was dynamically reduced by a factor of 0.1 every five epochs to ensure optimal convergence. A batch size of 32 was selected to balance computational efficiency and memory consumption.

To prevent overfitting, early stopping was implemented based on validation loss. The training process was conducted using PyTorch on GPU-enabled hardware to ensure efficient computation and faster convergence.

This structured approach ensured a fair evaluation of both models while addressing potential issues such as overfitting and unstable convergence, contributing to a reliable comparative analysis.

B. Dataset and Evaluation Metrics

The CXR 14 dataset [5], released by the NIH, contains 112,120 frontal CXRs from 30,805 patients. Each image is annotated for 14 disease categories, such as pneumonia, tuberculosis, and lung cancer. With its large and diverse dataset, it is ideal for multi-label classification tasks.

Preprocessing steps were applied to standardize the images, resizing them to 224×224 pixels for compatibility with both ViT and DenseNet models. Pixel normalization scaled the image pixel values to the range $[0, 1]$, and data augmentation techniques, including random flips, rotations, and zooms, enhanced model generalization and reduced overfitting.

To better understand the dataset's demographic composition, Figure 3 presents an exploratory comparison of patient age and gender density, illustrating the distribution differences between male and female patients. Such insights are crucial for evaluating potential biases in model predictions.

The dataset was annotated for 14 disease categories, which were treated as a multi-label classification task. Each image could belong to multiple classes, and binary cross-entropy was used as the loss function. Table II summarizes the disease distribution in the dataset.

C. Implementation - algorithm

As shown in Algorithm 1, the input image (step 1) is resized and divided into non-overlapping patches of fixed size (e.g., 16×16 pixels). Each patch is then flattened into a 1D vector. This approach allows the transformer model to treat image patches as tokenized elements similar to words in NLP. The flattened patch vectors are projected into a D-dimensional embedding

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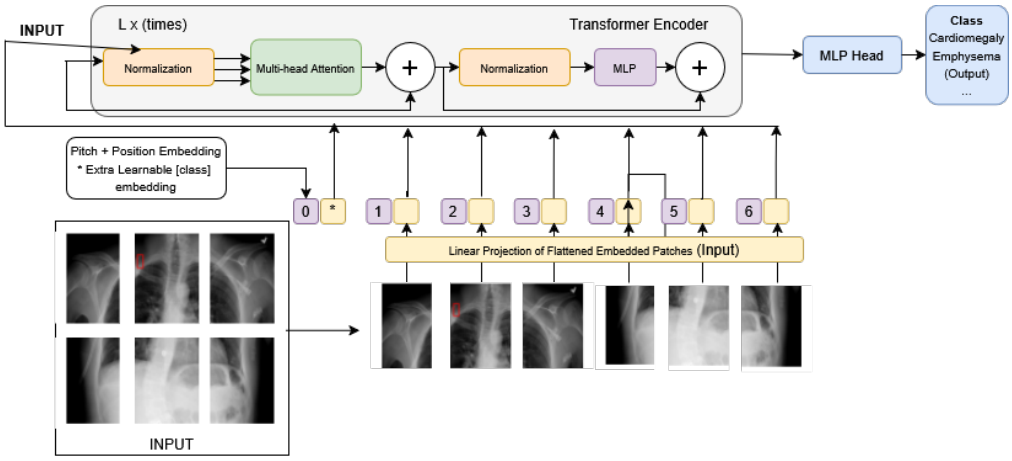


Fig. 2: ViT architecture for CXR analysis: it segments input images into patches, encodes positional information, and processes them through multi-head self-attention layers for multi-label classification.

Algorithm 1 ViT Algorithm for image classification

Require: Input image $x \in \mathbb{R}^{H \times W \times C}$
Ensure: Predicted labels for multi-label classification
Step 1: Image Patching
Resize input image to (H, W) and divide it into $(P \times P)$ patches
Flatten each patch into a 1D vector of size $P^2 \times C$
Step 2: Patch Embedding
Project each patch to a D-dimensional embedding space
Form $X_{\text{patch}} \in \mathbb{R}^{N \times D}$, where $N = \frac{H \times W}{P^2}$
Step 3: Adding Positional Encodings
Generate positional encoding $E_{\text{pos}} \in \mathbb{R}^{N \times D}$
Add to patch embeddings: $Z = X_{\text{patch}} + E_{\text{pos}}$
Step 4: Transformer Encoder
for each encoder layer l **do**
Compute multi-head self-attention: $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$
Apply Layer Normalization, Residual Connection, and Feedforward Network
end for
Step 5: Classification Head
Extract the class token's output and apply a fully connected layer with sigmoid activation
Step 6: Output Prediction
Generate final class probabilities =0

TABLE II: Disease distribution in the CXR 14 dataset

Disease	Description	# Images
Atelectasis	Collapse or closure of a lung (causes reduced or absent gas exchange)	15,000
Cardiomegaly	Enlargement of the heart (due to heart failure)	8,000
Effusion	Accumulation of fluid (in the pleural cavity around the lungs)	10,000
Infiltration	Abnormal material in the lung parenchyma (indicates infection or edema)	20,000
Mass	Localized swelling or lump in the lungs (both benign or malignant)	8,000
Nodule	Small, rounded growth in the lung tissue	7,500
Pneumonia	Inflammation of the lung tissue (by bacterial or viral infection)	5,000
Pneumothorax	Air or gas in the pleural space (causes lung collapse)	3,000
Consolidation	Region of lung tissue filled with liquid instead of air (seen in pneumonia)	2,000
Edema	Swelling caused by fluid accumulation (in the lungs from heart failure)	4,000
Emphysema	Condition where the alveoli (air sacs) are damaged (caused by smoking)	2,500
Fibrosis	Scarring of the lung tissue (due to long-term disease or exposure)	1,500
Pleural Thickening	Thickening of pleura (lining of the lungs) (due to infections or inflammation)	2,000
Hernia	Protrusion of an organ through a weak point (of chest cavity wall)	1,000

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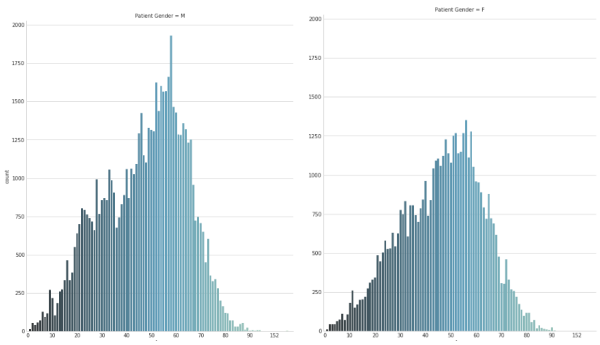


Fig. 3: Exploratory comparison of patient age and gender density: male (left) and female (right).

space (step 2). This transformation ensures consistent feature dimensions across all image patches. In step 3, positional encodings are added to the patch embeddings to retain spatial information. Without positional encodings, the transformer would lose track of spatial structure since self-attention is permutation-invariant. Multiple transformer encoder layers are stacked, each containing multi-head self-attention and feedforward layers (step 3). The self-attention mechanism allows the model to capture both local and global dependencies within the image. In Step 5, the class token's output vector serves as a global representation of the entire image. This token is passed through a fully connected layer with sigmoid activation for multi-label classification tasks. Finally, Step 6 generates the output probabilities for each class, providing the final model prediction.

D. Our results, comparison, and discussions

By correlating parameter counts with performance (see Table III), one could optimize model selection, balancing accuracy, efficiency, and resource constraints.The results highlight the complementary strengths of ViT and DenseNet in CXR analysis. ViT leverages self-attention to capture global dependencies, excelling in diseases like pneumothorax and pneumonia that require broader contextual understanding. DenseNet, with its efficient feature reuse and dense connectivity, is well-suited for recognizing localized features such as nodules and masses. Despite ViT's higher computational cost, its interpretability through attention maps is crucial for medical applications, while DenseNet's lower inference time makes it practical for real-time or resource-constrained settings.

A detailed comparison of model complexities aids selection. ViT-B16 has approximately 86M parameters due to its global self-attention, enhancing performance but increasing computational demands. DenseNet-121, with around 8M parameters, efficiently reuses features, maintaining accuracy in localized tasks. ResNet-50 (25.6M parameters) balances size and performance through efficient feature extraction. EfficientNet-B7 (66M parameters) achieves high accuracy via optimized scaling, while Xception (22.9M parameters) employs depthwise separable convolutions to reduce computational overhead. As shown in Table IV, ViT outperforms ResNet50 and Xception in AUC and accuracy while maintaining comparable F1-scores.

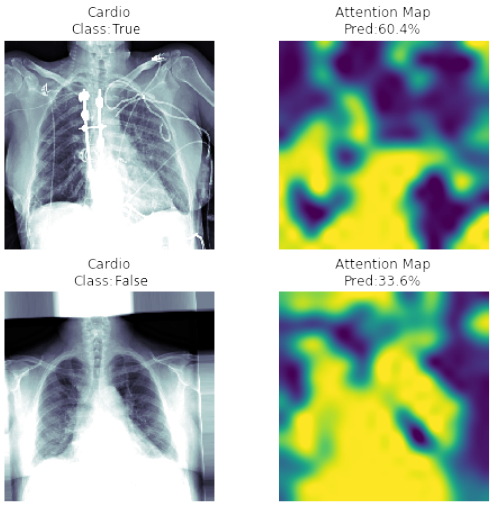


Fig. 4: Attention map highlighting ROIs used by the ViT model to identify abnormalities in CXRs.

Our analysis underscores that ViT, though computationally intensive, enhances interpretability through attention maps, offering a global contextual understanding beneficial for complex medical imaging. In contrast, DenseNet's efficiency and faster inference times make it a strong baseline for resource-constrained environments. The results in Table IV reinforce ViT's superior performance, particularly in AUC and accuracy, making it a promising choice for medical imaging tasks requiring deeper contextual awareness.

IV. Conclusion and future work

In this paper, we have studied how ViT excel in analyzing CXRs by leveraging global attention mechanisms to model long-range dependencies across entire images. Our study highlights that ViT consistently outperform DenseNet in detecting pathologies requiring an understanding of interdependent features across larger image regions, such as Emphysema and Cardiomegaly. Future work could explore hybrid models that integrate the strengths of ViT and CNNs to balance interpretability and computational efficiency.

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TABLE III: Results: DenseNet-121 and ViT for multi-Label CXR classification

Pathology	DenseNet AUC	ViT AUC	DenseNet Accuracy	ViT Accuracy	DenseNet F1-Score	ViT F1-Score
Cardiomegaly	0.9126	0.9150	0.8331	0.8345	0.1804	0.1835
Emphysema	0.9360	0.9380	0.8767	0.8780	0.2484	0.2520
Effusion	0.9046	0.9075	0.7935	0.7960	0.5150	0.5185
Hernia	0.8675	0.8700	0.7343	0.7375	0.0087	0.0091
Infiltration	0.6655	0.6675	0.6708	0.6730	0.3943	0.3960
Mass	0.8861	0.8880	0.8353	0.8360	0.3040	0.3070
Nodule	0.7899	0.7915	0.7551	0.7565	0.2437	0.2460
Atelectasis	0.8338	0.8365	0.7447	0.7460	0.3675	0.3705
Pneumothorax	0.9061	0.9070	0.7795	0.7800	0.3252	0.3280
Pleural Thickening	0.8148	0.8170	0.6859	0.6880	0.1325	0.1350
Pneumonia	0.8067	0.8100	0.8032	0.8045	0.0677	0.0705
Fibrosis	0.7861	0.7890	0.7192	0.7210	0.0350	0.0375
Edema	0.9144	0.9155	0.8062	0.8080	0.1325	0.1355
Consolidation	0.8178	0.8200	0.7443	0.7455	0.1798	0.1820

TABLE IV: Comparison: ViT model and benchmark works on NIH CXR 14 dataset for cardiomegaly disease

Model	AUC	Accuracy	F1-score
DenseNet121 [10]	0.9126	0.8345	0.1835
ResNet50 [18]	0.8900	0.8200	0.1750
EfficientNetB7 [19]	0.9100	0.8300	0.1800
Xception [20]	0.9060	0.8245	0.1820
Vision Transformer	**0.9150**	**0.8345**	**0.1835**

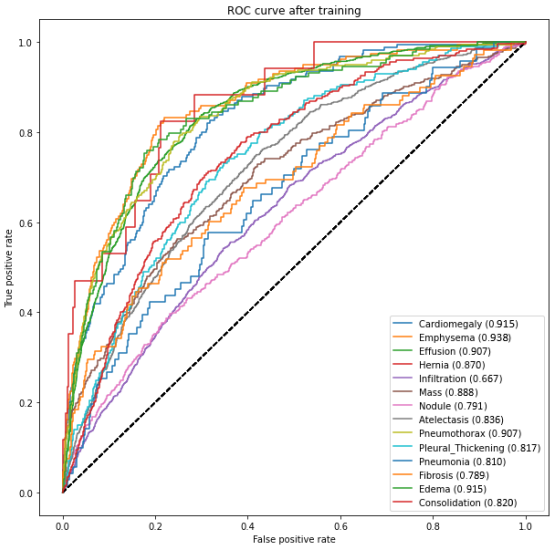


Fig. 5: ROC curves for the proposed method using multi-label CXR classification for different disease types.

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